Correlation between economic, health, and institutional context of applicants' country and INZ student visa approval decisions

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Abstract. This article examines country-level variation in Immigration New Zealand fee-paying student visa approval rates for the 2022–2025 period. The study combines publicly available INZ decision data with publicly available country-level reference variables from the World Bank, the World Health Organization, and UN M49 regional classifications. The analysis uses descriptive statistics, Pearson correlations, linear regression approximations, alternative regional specifications, residual diagnostics, and a weighted robustness check based on log-transformed application volume.

The results show that approval rates vary systematically across applicant nationalities and are associated with application volume, income classification, regional classification, and broader country-level development indicators. In the modelling table, several health, income, internet-use, mortality, and life-expectancy indicators are strongly correlated with approval rates, although these indicators also overlap substantially with one another. Among the tested regression specifications, the UN M49 regional model has the strongest fit, followed by the World Bank regional model, the WHO regional model, and the no-region model. A weighted UN M49 experiment gives similar overall fit but weakens several individual regional coefficients, supporting the model-fit conclusion while limiting interpretation of particular coefficients.

The findings should be interpreted as country-level descriptive evidence, not as causal evidence about individual visa decisions. The analysis is explicitly based on INZ's published visa instructions, which set out common formal requirements for student visa applicants, including evidence of enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character requirements. The observed country-level differences are therefore not interpreted as evidence of officer behaviour, discriminatory intent, bias, or unequal treatment in individual application assessment. Rather, they are treated as aggregate descriptive outcomes that may reflect country-level differences in documentary, financial, institutional, and verification conditions under the same formal decision framework. The principal limitation is that the data are aggregated by country and do not include applicant-level records, programme choice, education provider information, documentation quality, or detailed processing factors. The study identifies a reproducible empirical structure and points to further research on other receiving countries, additional country-level variables, and formal inter-country education or migration relationships.

Keywords: student visas · Immigration New Zealand · country-level analysis · approval rates · regional classification

1 Introduction

International student mobility is now a large but unevenly distributed part of higher education. UNESCO reports that 6.9 million higher-education students study outside their home country, more than three times the number in 2000 [33]. New Zealand is a small destination in this global system, but not a marginal case for its own economy. In 2024 it had 50,300 international tertiary learners, approximately 0.7 percent of the global internationally mobile higher-education population if compared with UNESCO's 6.9 million figure [26, 33]. Across all education sectors, New Zealand recorded 83,425 international student enrolments in 2024 [8]. The Government's international education plan identifies the sector as a major export activity and aims to increase its value from NZ$3.6 billion in 2024 to NZ$7.2 billion by 2034 [25].

New Zealand is therefore a useful empirical setting for studying student visa decisions. It is geographically isolated, relatively small, and economically interested in international education, so student entry is both commercially important and administratively filtered. A fee-paying student visa application is not a casual expression of interest. INZ's published instructions require evidence and documents concerning enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character matters [16]. Appendix 10.2 summarises the preparation steps that make these applications document-intensive. The process is costly and document-intensive, so the observed applications are unlikely to be dominated by low-effort or "noise" submissions. Applicants usually need to have already assembled institutional, financial, and personal evidence before a decision appears in the published visa statistics.

The empirical problem is that approval rates differ by applicant nationality, but the reasons for these differences are not visible in the public INZ decision tables. The published rules used in this article state common formal requirements; they do not state separate approval rules by nationality. For this reason, observed country-level differences should not be treated as direct evidence of unequal treatment in individual assessment. They are better framed as an aggregate empirical pattern that may reflect differences in application volume, documentation conditions, financial evidence, institutional context, verification conditions, or other country-level circumstances under a common formal decision framework.

The research question is narrow: are country-level INZ fee-paying student visa approval rates for 2022–2025 associated with application volume, national development and health indicators, income classification, and alternative regional classifications? The article answers this question by joining public INZ decision data to public World Bank, WHO, and UN M49 reference data and by comparing descriptive statistics, correlations, and regression approximations across alternative regional specifications.

Previous quantitative research on international student migration has used visa refusal or visa-policy data together with student mobility or enrolment outcomes, particularly in the United States and Canada. Chen examines U.S. F-1 visa restrictiveness and international student enrolment, while Gonnot studies Canadian student visa policy and community-college enrolment [6, 9]. Other work models international student mobility using origin- and destination-country characteristics, including World Bank development indicators. However, peer-reviewed work directly modelling New Zealand student visa approval rates appears limited, and studies combining country-level student visa decision data with both World Bank development indicators and WHO health indicators are also uncommon. This article addresses that gap with a reproducible country-level descriptive design.

2 Literature Review

Some are born to Sweet Delight,
Some are born to Endless Night [4].

International education is often presented as an open global market in which students choose programmes and institutions compete for talent. This framing hides a basic constraint: students cannot simply choose to study abroad unless a state permits them to enter and remain. Cross-border education is therefore not only an academic or economic process; it is also a migration process governed by visa systems. Sassen notes that international law does not provide a general right to enter foreign territory [30]. Shachar similarly emphasises that the practical value of citizenship is closely tied to mobility rights attached to passports [31]. Visa systems therefore sit at the boundary between education markets and state control over entry.

This matters because mobility is unevenly distributed. Neumayer argues that states use visa restrictions to manage a trade-off between facilitating entry for some passport holders and deterring others for reasons of security, immigration control, economic interest, and political interest [24]. Mau develops a related account of the global mobility divide, in which visa policies produce unequal access to cross-border movement [23]. In this literature, visa policy is not a neutral background condition. It is one of the central mechanisms through which access to international mobility is facilitated for some groups and constrained for others.

International student mobility is part of this wider mobility system. Oduwaye reviews the challenges faced by international students and shows that financial stability and other adaptation conditions remain central issues across host countries [27]. Gulel describes visa inequities in academic mobility and treats visa access as a form of privilege rather than a purely administrative step [10]. These studies support the premise that international education cannot be analysed only through enrolment demand or university admission. For many applicants, admission to a programme is only one step in a longer chain of documentary, financial, institutional, and immigration requirements.

The New Zealand case is relevant because international education is a policy and economic sector, while student entry remains governed by formal visa requirements. Craymer reports New Zealand's policy aim to expand the international education market substantially by 2034 [7]. At the same time, Spoonley discusses contemporary immigration policy challenges in New Zealand, and Bonnett reports concerns about Immigration New Zealand's processing systems and risk capacity [32, 5]. INZ's published student visa instructions require applicants to provide evidence and documents, including enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character information [16]. Appendix 10.2 summarises the preparation steps that make these applications document-intensive. These formal requirements make the student visa decision process a suitable empirical setting for studying how country-level conditions may appear in aggregate approval-rate patterns.

Existing quantitative migration research provides methodological precedents for joining migration-related outcomes with country-level indicators. Weber examines the relationship between development and international student migration, showing that student mobility can be studied as part of broader development and migration transitions [35]. Bertoli uses international migration data joined with macroeconomic sources and highlights the importance of modelling migration in relation to multiple country-level contexts [2]. Abbott applies ordinary least squares regression in a gravity-model framework for international student migration and uses country-level variables to explain student flows [1]. These studies differ from the present analysis in outcome and design, but they support the use of country-level explanatory variables and regression-based approximation for migration-related questions.

Prior research also shows that visa policy and approval patterns can shape international student mobility. Chen studies U.S. F-1 visa restrictiveness and international student enrolment, while Gonnot examines Canadian student visa policy and international student migration [6, 9]. Gonnot is especially relevant because it links lower study-visa approval rates for applicants from developing countries to information and documentation problems that may fail to resolve immigration officers' concerns [9]. This supports the present article's cautious interpretation: lower approval rates need not be read as evidence of unequal person-to-person treatment; they may reflect differences in the evidence available to satisfy formal visa requirements.

Country-level and regional explanations are also supported by the broader student-mobility literature. Perkins and Neumayer, Abbott, Bessey, and Lanati and Thiele use country-level data to analyse determinants of international student flows, including geographic factors, networks, institutional quality, political freedom, GDP per capita, fees, and foreign aid projects [29, 1, 3, 22]. This literature justifies treating student mobility and visa outcomes as phenomena that can be associated with country-level conditions, while also showing that such models cannot identify all applicant-level mechanisms.

Some related research examines unequal outcomes in labour-market or social settings. Hersch studies the relationship between skin colour and immigrant labour-market outcomes in the United States, and Oreopoulos uses a field experiment to examine discrimination against applicants with foreign experience or non-English names [11, 28]. These studies are not direct models of INZ student visa decisions and are not used here to infer bias in visa assessment. Their relevance is narrower: they show that observed outcomes involving migrants can vary by origin-linked characteristics, and that empirical work must be careful to distinguish documented unequal treatment from aggregate differences that may arise through other mechanisms.

The present article contributes to this literature by analysing publicly available INZ fee-paying student visa decision data at country level and joining it with publicly available World Bank, WHO, and UN M49 reference data. Previous studies have used visa restrictiveness, student mobility, enrolment, or country-level development variables, but peer-reviewed work directly modelling country-level student visa approval rates with both development and health indicators appears limited. The analysis therefore addresses a specific empirical gap: whether country-level approval-rate variation in New Zealand student visa decisions is associated with application volume, national development indicators, income classification, and alternative regional classifications.

3 Data

The primary source data are publicly available Immigration New Zealand offshore or overseas student visa decision tables for 2022, 2023, 2024, and 2025 [12–15]. The project repository is available on GitHub [18], and its internal structure is described in Appendix 10.1. The INZ decision tables report fee-paying student visa decisions by applicant nationality or country. The yearly tables use slightly different column names, but the common information is the country or nationality, number approved, number declined, total or volume of applications, approval rate, and decline rate.

The unit of observation in the prepared INZ data is a country or applicant nationality aggregated across the four-year period. The prepared INZ data contain 179 countries or applicant nationalities. For each country or applicant nationality, the data report total approvals, total declines, total applications, approval rate, and the base-10 logarithm of total applications. Approval rate is calculated as approved applications divided by total applications. These fields are used for descriptive results on application volume and overall approval rates.

The modelling dataset joins the prepared INZ table to publicly available country-level reference data from the World Bank, the World Health Organization, and UN M49 [37, 36, 38, 39, 34]. The World Bank reference table contributes FY2026 region, income group, lending category, GDP per capita, GNI per capita using the Atlas method, life expectancy, internet-use percentage, and urban-population percentage. The WHO reference table contributes WHO region, under-five mortality, maternal mortality, DTP3 immunization coverage, and physician density. The UN M49 reference table contributes an alternative regional classification.

Country names are harmonised across the INZ, World Bank, WHO, and UN M49 data before modelling. The combined modelling data contain 110 countries after country matching, removal of rows with missing numerical values needed for the models, and exclusion of countries with fewer than 10 total applications across 2022–2025. Countries present in the prepared INZ data but absent from the regression data are reported separately in Appendix Table 16.

Table 1 summarises the fields in the combined modelling data. For string fields, only the number of unique values is reported. For numeric fields, the table reports the minimum, maximum, number of unique values, mean, and sample standard deviation.

Table 1: Statistical characteristics of the combined modelling data

Column	Type	Min.	Max.	Unique	Mean	SD
Country	String	–	–	110	–	–
`WHO_Region`	String	–	–	6	–	–
`WB_FY2026_Region`	String	–	–	7	–	–
`WB_FY2026_Income_Group`	String	–	–	5	–	–
`WB_FY2026_Lending_Category`	String	–	–	4	–	–
UN M49	String	–	–	21	–	–
Approval rate	Numeric	0.000	1.000	100	0.754	0.248
`Applications_log10`	Numeric	1.041	4.550	92	2.201	0.857
`WHO_Under5_Mortality_per_1000_live_births`	Numeric	2.069	104.9	110	19.42	19.23
`WHO_Maternal_Mortality_per_100000_live_births`	Numeric	1.096	992.8	110	86.94	142.1
`WHO_Physician_Density_per_10000_population`	Numeric	0.610	65.76	109	22.07	17.02
`WHO_DTP3_Immunization_Coverage_pct`	Numeric	39.00	99.00	33	87.47	13.32
`GDP_per_capita_current_USD`	Numeric	413.8	137,781.7	110	20,242.7	26,831.7
`Life_expectancy_at_birth_years`	Numeric	54.64	84.58	107	74.85	6.456
`Internet_users_pct_population`	Numeric	8.950	100.0	108	74.20	25.91
`Urban_population_pct`	Numeric	15.41	100.0	109	63.36	22.27
`GNI_per_capita_Atlas_current_USD`	Numeric	370.0	98,170.0	106	18,910.2	23,703.6

4 Methods

4.1 Computational Workflow

The original computation and model evaluation were developed in KNIME Analytics Platform [21]. The KNIME workflows were then exported into Python scripts and notebooks using the open-source `knime2py` utility [20], available through `k2pweb.org` [19]. The exported Python code provides the text-based computational record and is the source used for the final reported results. The KNIME workflows, exported Python files, and supporting project sources are available in the GitHub repository [18].

The workflow first harmonises the yearly INZ decision tables for 2022, 2023, 2024, and 2025. Differences in column names are reconciled, common decision fields are retained, and the yearly tables are concatenated and aggregated by country or applicant nationality. The aggregated INZ table records total approvals, total declines, total applications, approval rate, and the base-10 logarithm of total applications.

The aggregated INZ data are then joined to the World Bank, WHO, and UN M49 reference data after country-name harmonisation. During this stage, unusable columns are removed, missing string categories are filled where this preserves interpretable category membership, and rows with missing numerical values required for modelling are removed. Countries with fewer than 10 total applications over the four-year period are also excluded from the regression table.

The modelling stage uses the combined country-level table produced by this process. Numerical explanatory variables are min–max normalised before fitting. Categorical predictors, including income, lending-category, and regional variables, are represented as indicator variables with one category omitted as the reference level. Table 2 reports the reference levels used for these predictors. The dependent variable is country-level approval rate. The workflow then exports coefficient tables, fitted values, prediction-rate summaries, model scores, residual diagnostics, and robustness summaries used in the Results and sensitivity-check sections.

Table 2. Reference levels for categorical predictors

Categorical predictor	Omitted reference level
World Bank income group	High income
World Bank lending category	Blend
UN M49 regional model	Caribbean
World Bank regional model	East Asia & Pacific
WHO regional model	African Region

4.2 Descriptive Statistics and Regression

The first stage is descriptive. The unified INZ table is used to report the number of countries or applicant nationalities, total applications, approval rates, and the distribution of application volume. Application volume is represented both as total applications and as the base-10 logarithm of total applications, because application counts are highly uneven across countries.

Associations between approval rate and individual country-level variables are examined with Pearson correlation coefficients. These correlations are reported with the correlation coefficient, p-value, and degrees of freedom where relevant. They are used for the relationship between approval rate and application volume, and for relationships between approval rate and country-level indicators such as mortality, life expectancy, internet use, physician density, GNI per capita, and GDP per capita.

The main modelling step uses ordinary least squares linear regression to approximate country-level approval rate as a function of selected WHO and World Bank indicators, income and lending categories, and, in three specifications, regional classification variables. The statistical procedures used here are standard tools in applied regression and statistical learning: correlation, linear regression, inference with p-values, model assessment, diagnostic inspection, transformations, categorical predictors, overfitting concerns, and cautious interpretation of fitted models [17]. This modelling choice also follows related quantitative migration research at a general methodological level. Weber and Van Mol use development indicators in a gravity-model analysis of international student migration, showing that student mobility can be modelled in relation to country development conditions [35]. Bertoli and Fernández-Huertas Moraga join migration data with IMF and World Bank macroeconomic data, providing a precedent for combining migration outcomes with external country-level reference data [2]. Abbott and Silles apply ordinary least squares within a log-linear gravity-model framework for international student migration and interpret country-level predictors, income variables, R^2, and coefficient significance in that setting [1]. The present models are not gravity models and do not analyse student flows; they use a comparable regression-based strategy to approximate country-level visa approval rates.

The models are explanatory approximations. They are not operational prediction systems and are not causal designs. Their purpose is to measure how much of the observed country-level variation can be fitted by the current set of country-level variables.

4.3 Regional Specifications

The country-level numerical variables and the regional variables have different methodological status. Health, income, internet-use, mortality, and life-expectancy measures are treated as country-level reference indicators. For a given source and year, these variables have a defined value for a country. Regional classification is different. A region is an external grouping imposed on countries, and the same country can be assigned to different regional systems depending on the classification scheme. For this reason, regional division is treated as a modelling choice rather than as an inherent property of the country.

The analysis estimates four related specifications. Three models use alternative external regional classifications: UN M49 regions, World Bank FY2026 regions, and WHO regions. A fourth model excludes regional variables. The same dependent variable and the same non-regional country-level predictors are used across the specifications, so the comparison focuses on whether the choice of regional grouping changes model fit. This design separates two questions: whether country-level reference indicators are associated with approval rates, and whether an external regional classification adds explanatory structure beyond those indicators.

4.4 Model Evaluation and Residuals

The regression models are evaluated with standard linear model fit statistics. The main comparison across model specifications uses R-squared, mean absolute error, mean squared error, and root mean squared error. These statistics are used to compare the UN M49, World Bank regional, WHO regional, and no-region specifications. They measure fit within the current country-level modelling table and should not be read as validation of a causal mechanism.

Individual regression terms are assessed with coefficient estimates and p-values. The p-values are used to identify variables and regional indicators that cross the conventional $p < 0.05$ threshold within each specification. Because the models include correlated country-level indicators and categorical regional variables, statistical significance is interpreted as conditional association within the fitted model rather than as evidence of an independent causal effect.

Prediction diagnostics are based on residuals and relative fitted error. Residual summaries include the mean residual, residual standard deviation, mean absolute residual, and maximum absolute residual. Residuals are also inspected against fitted values to check whether unexplained error is concentrated in particular fitted-value ranges. Relative residuals are examined with `Prediction_rate`, defined as the ratio of observed approval rate to fitted approval rate. Countries with observed approval rates below 80 percent or above 120 percent of the fitted value are treated as large proportional deviations for descriptive comparison. These cases are not treated automatically as technical outliers; they identify countries where the current variables fit observed approval rates poorly.

4.5 Robustness and Interpretation

Robustness checks compare the same quantities across alternative regional classifications and a weighted UN M49 specification. The weighted model uses the base-10 logarithm of total applications as the observation weight, giving more influence to countries with larger log-transformed application volumes. It is evaluated with weighted R-squared, weighted mean absolute error, weighted mean squared error, and weighted root mean squared error, and also with unweighted fit statistics for comparison with the main UN M49 model.

The interpretation of these statistics rests on INZ's published student visa instructions, which define common formal requirements for applicants. The models therefore do not use country-level differences to infer unequal treatment, bias, or discriminatory decision-making. They use those differences to study variation in the conditions under which applicants from different countries prepare, document, finance, and verify applications within the same formal decision framework. The analysis is also limited by the aggregation of applications to country level and by overlap among country-level predictors. The correlation output is therefore used not only as descriptive evidence but also as a warning that individual regression coefficients should not be overinterpreted as independent effects.

5 Results

5.1 Descriptive Results

The unified INZ decision table contains 179 countries or applicant nationalities and 148,746 total fee-paying student visa applications across 2022–2025. Across these records, the overall approval rate is approximately 81.3 percent.

Application volume is unevenly distributed across countries. The largest application counts are from China, India, Japan, Nepal, the United States of America, Sri Lanka, Germany, Thailand, the Philippines, and South Korea. These high-volume countries do not all have the same approval rate. For example, China, Japan, the United States of America, Germany, and South Korea have approval rates above 96 percent, while India and Nepal have lower approval rates in the current prepared table.

The relationship between application volume and approval rate is positive but not strong. In the unified country-level table, the Pearson correlation between log10 total applications and approval rate is about 0.27, with $p = 0.000254$ and 177 degrees of freedom. This supports a limited descriptive statement: countries with more applications tend to have higher approval rates, but application volume alone does not explain most of the variation in approval rates. Figure 1 shows this relationship graphically.

5.2 Correlation With Country-Level Indicators

After joining the INZ data with WHO, World Bank, and UN M49 reference data, the combined modelling table contains 110 countries. This reduction reflects the availability of matched country-level reference variables, the removal of rows with missing numerical values, and the exclusion of countries with fewer than 10 total applications across the four-year period. The countries excluded before regression modelling are listed in Appendix Table 16.

Several country-level indicators are strongly correlated with approval rate. The direction of the correlations is consistent with a broad development and administrative-capacity pattern. Approval rates are lower where mortality indicators are higher, and higher where life expectancy, internet use, physician density, and income indicators are higher. Table 3 reports the strongest bivariate correlations used in this interpretation.

The same correlation output also shows that several explanatory variables are strongly correlated with each other. For example, WHO under-five mortality is strongly negatively correlated with life expectancy ($r = -0.888$) and internet-user percentage ($r = -0.770$), and strongly positively correlated with WHO maternal mortality ($r = 0.881$). This means that the numerical country-level indicators partly measure overlapping dimensions of national development, health conditions, and institutional capacity.

These correlations indicate that approval rates are associated with broad country-level development, health, and income measures. They should not be

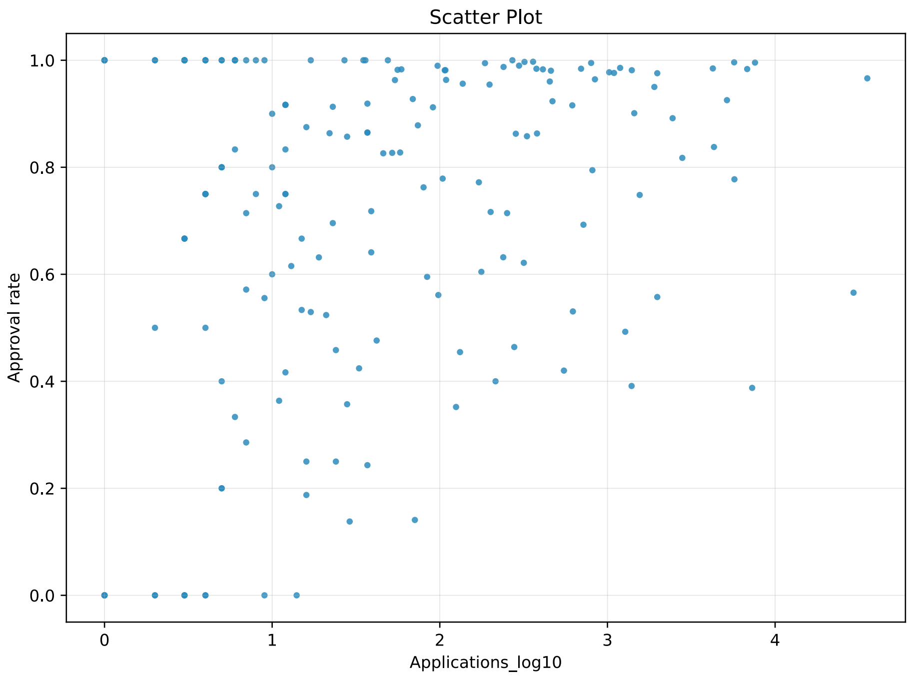

Fig. 1. Total applications, log10 transformed, and student visa approval rate by country or applicant nationality. The PDF figure is generated from `data/results/applicationslog10_to_approval_rate.svg`.

Table 3. Correlations between approval rate and selected country-level indicators

Indicator	r	p
WHO under-five mortality per 1,000 live births	-0.746	$1.24e{-30}$
WHO maternal mortality per 100,000 live births	-0.681	$7.97e{-24}$
Life expectancy at birth	0.715	$4.19e{-27}$
Internet users as percentage of population	0.633	$7.62e{-20}$
Physician density per 10,000 population	0.552	$1.56e{-14}$
GNI per capita, Atlas method	0.520	$8.18e{-13}$
GDP per capita, current USD	0.501	$7.42e{-12}$

read as causal effects. The data are aggregated by country, and the variables describe country-level conditions rather than individual applicants.

This interpretation is limited to formal and administrative conditions surrounding applications. The analysis is explicitly based on INZ's published visa instructions, which set out common formal requirements for student visa applicants, including evidence of enrolment, tuition arrangements, sufficient funds, bona fide temporary-entry intent, and relevant health and character requirements. The observed country-level differences are therefore not interpreted as evidence of officer behaviour, discriminatory intent, bias, or unequal treatment in individual application assessment. Rather, they are treated as aggregate descriptive outcomes that may reflect country-level differences in documentary, financial, institutional, and verification conditions under the same formal decision framework.

5.3 Regression Approximation

Four linear regression specifications were estimated. Three specifications use different regional classifications: UN M49, World Bank regions, and WHO regions. A fourth specification excludes regional classification variables. In each case, approval rate is the dependent variable, and the model uses selected WHO and World Bank country-level variables, with or without regional classification variables, as explanatory inputs.

Table 4. Regional classification specifications used in the regression models

Specification	Number of regions in modelling table
UN M49 regional classification	21
World Bank FY2026 regional classification	7
WHO regional classification	6

The UN M49 specification has the strongest fit among the current models, while the no-region specification has the weakest fit. Table 5 reports the fit statistics for all four specifications.

Table 5. Model fit by regional specification

Specification	R-squared	MAE	RMSE
UN M49 regional model	0.867	0.069	0.090
World Bank regional model	0.812	0.082	0.107
WHO regional model	0.786	0.087	0.114
No-region model	0.706	0.100	0.134

These scores suggest that the current country-level variables explain a substantial share of the variation in country-level approval rates, and that regional classification adds explanatory structure to the model. The no-region model still fits a large part of the variation, but all three regional specifications improve fit. The better fit of the UN M49 model may reflect the more detailed regional division used in that specification. This interpretation is supported by the robustness and sensitivity checks described below, but it should remain limited to model fit rather than causal explanation.

The coefficient tables show that model fit is not produced by regional variables alone. Other variables from the combined dataset also contribute to the fitted approval rates. The signs and magnitudes of these coefficients should be interpreted carefully because the model includes correlated country-level indicators and categorical variables. The coefficients show conditional associations within the fitted model, not independent causal effects.

In the UN M49 experiment, only a small set of non-region variables and regional indicators have p-values below 0.05. Table 6 lists these terms. In this specification, life expectancy remains statistically visible after regional controls are included, while most other numerical country-level indicators do not.

Table 6. Terms below $p < 0.05$ in the UN M49 specification

Term	Type	Coefficient	p
Urban population percentage	Variable	-0.279	0.0053
Life expectancy at birth	Variable	0.499	0.0128
World Bank lending category: missing	Category	0.242	0.0156
Northern Africa	Region	-0.254	0.0109
Middle Africa	Region	-0.275	0.0219
Southern Asia	Region	-0.210	0.0336

The missing category for World Bank lending category needs a specific interpretation. In the modelling table, all countries in this category are high-income countries. They include countries such as Japan, Singapore, South Korea, Canada, the United States of America, Germany, France, Great Britain, Switzerland, Norway, Sweden, Austria, Belgium, Denmark, Ireland, Italy, Spain, Kuwait, Oman, Saudi Arabia, and the United Arab Emirates. The category therefore should not be read as ordinary data loss. In this dataset, it mainly identifies high-income economies without an assigned World Bank lending category. Its positive coefficient is consistent with the generally high approval rates observed for this group, but it should be interpreted as a proxy for this country group rather than as an effect of missing information.

The negative regional coefficients in Table 6 also require a categorical interpretation. Regional variables are entered as indicator variables, so each regional coefficient is estimated relative to the omitted reference region, not relative to zero approval or to the global average. A negative coefficient therefore means that, conditional on the other variables in the model, the fitted approval rate for that region is lower than the fitted approval rate for the reference region. It does not mean that the region has a negative approval rate, and it should not be read as an absolute ranking of regions.

The World Bank regional experiment combines income-group effects, urban population percentage, lending-category information, and broad World Bank regional effects. Table 7 lists the terms below $p < 0.05$ in this specification.

Table 7. Terms below $p < 0.05$ in the World Bank regional specification

Term	Type	Coefficient	p
World Bank low-income status	Category	-0.407	$5.48e{-}05$
World Bank lower-middle-income status	Category	-0.136	0.0461
World Bank lending category: missing	Category	0.224	0.0171
Urban population percentage	Variable	-0.255	0.0061
Middle East, North Africa, Afghanistan and Pakistan	Region	-0.270	$8.19e{-}08$
Europe and Central Asia	Region	-0.185	0.000369
South Asia	Region	-0.232	0.000555
Sub-Saharan Africa	Region	-0.156	0.0042

The WHO regional experiment has fewer significant regional terms than the World Bank regional model and a weaker overall fit. Table 8 lists the terms below $p < 0.05$ in this specification. The Region of the Americas is close to the 0.05 threshold but does not cross it ($p = 0.0691$).

Table 8. Terms below $p < 0.05$ in the WHO regional specification

Term	Type	Coefficient	p
World Bank low-income status	Category	-0.442	$4.23e{-}05$
World Bank lower-middle-income status	Category	-0.178	0.0127
World Bank lending category: missing	Category	0.231	0.0198
Urban population percentage	Variable	-0.237	0.0132
Western Pacific Region	Region	0.198	0.000975

In the no-region experiment, the model has no regional indicators, so the fit depends only on non-region country-level variables. Table 9 lists the terms below $p < 0.05$ in this specification. These variables carry substantial explanatory information, even though the no-region model fits worse than all three regional specifications.

The full coefficient and p-value outputs for the four regression specifications are reported in Appendix Tables 17–20.

Table 9. Terms below $p < 0.05$ in the no-region specification

Term	Type	Coefficient	p
World Bank low-income status	Category	-0.609	$2.50e{-07}$
World Bank lower-middle-income status	Category	-0.224	0.00465
WHO DTP3 immunization coverage	Variable	-0.204	0.0152
Urban population percentage	Variable	-0.264	0.0081

5.4 Prediction Deviations

The prediction tables and fitted-value plots compare observed approval rates with model-fitted approval rates; the country-level observed and fitted values are reported in Appendix Table 24. In all four specifications, the mean observed approval rate and mean fitted approval rate are both approximately 0.754 in the modelling table. Figure 2 shows the same comparison graphically. The horizontal axis gives the observed approval rate, and the vertical axis gives the fitted approval rate. Points close to the diagonal represent countries where the model fits the observed approval rate closely; points far from the diagonal identify countries where the current country-level variables and regional classification leave more unexplained variation.

The plots support the model-comparison results reported above. The UN M49 specification produces the closest visual concentration around the fitted-observed diagonal, while the no-region specification leaves a visibly wider spread. The World Bank and WHO regional specifications fall between these two cases. This pattern is consistent with the score table: regional variables improve fit, and the more detailed UN M49 classification gives the strongest approximation among the current models.

The number of large proportional deviations differs by specification. Table 10 uses the rule that a country is outside the expected band when the observed approval rate is below 80 percent or above 120 percent of the fitted value. The UN M49 model has the fewest such countries, again suggesting that it gives the closest fit among the current specifications.

Table 10. Countries outside the 80–120 percent fitted-value band

Specification	Below 80%	Above 120%	Total
UN M49 regional model	10	5	15
World Bank regional model	11	8	19
WHO regional model	15	10	25
No-region model	15	13	28

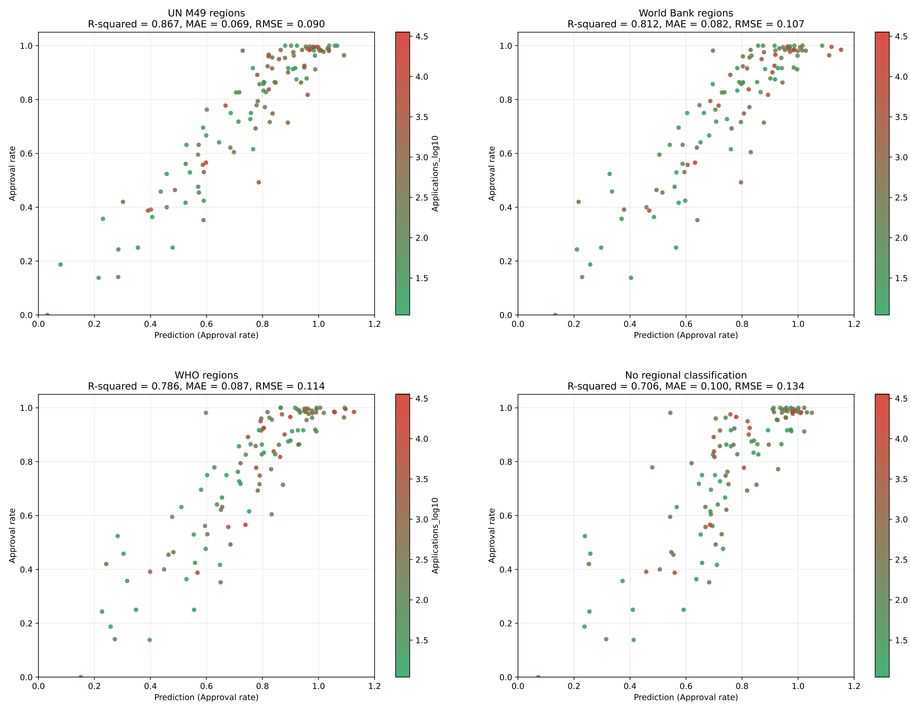

Fig. 2. Observed approval rate and fitted approval rate in the four regression specifications: UN M49 regions, World Bank regions, WHO regions, and no regional classification. The plots are generated from `data/results/approximation_sp_m49.svg`, `data/results/approximation_sp_wb.svg`, `data/results/approximation_sp_who.svg`, and `data/results/approximation_sp_no_region.svg`.

Table 11 lists the countries outside this band in the UN M49 model. These are the countries where the best-fitting current specification still leaves the largest proportional fitted-value deviations.

Table 11. Large proportional deviations in the UN M49 model

Country	Observed	Fitted	Ratio
Chad	0.000	0.032	0.000
Afghanistan	0.141	0.285	0.495
Uzbekistan	0.250	0.479	0.522
Turkey	0.352	0.589	0.597
Myanmar	0.493	0.786	0.626
Cameroon	0.138	0.215	0.642
Yemen	0.250	0.355	0.704
Tanzania	0.424	0.590	0.719
Algeria	0.417	0.525	0.794
Ghana	0.455	0.573	0.794
...			
Kazakhstan	0.762	0.601	1.269
Timor Leste	0.981	0.729	1.346
Nigeria	0.420	0.302	1.392
Sudan	0.357	0.230	1.551
Democratic Republic of Congo	0.188	0.079	2.375

These cases should not automatically be treated as technical outliers. They show where the current country-level variables and regional classifications do not fit the observed approval rate well. They may point to omitted variables, data quality issues, country-specific applicant composition, policy or processing factors, or other mechanisms not represented in the current dataset.

6 Robustness or Sensitivity Checks

The first sensitivity check is already built into the current workflow: the same modelling task is estimated with three alternative regional classifications and with no regional classification. As shown in Table 5, the UN M49 model has the strongest fit, followed by the World Bank regional model, the WHO regional model, and the no-region model. This comparison supports the statement that regional classification matters for model fit and that the more detailed UN M49 classification fits the current data best.

The score tables support the same ordering across several metrics. R-squared declines from 0.867 in the UN M49 model to 0.812 in the World Bank regional model, 0.786 in the WHO regional model, and 0.706 in the no-region model.

Mean absolute error and root mean squared error move in the opposite direction: they are smallest in the UN M49 model and largest in the no-region model.

The prediction-rate tables provide a second check. Table 10 shows that the UN M49 model produces the fewest countries outside the 80–120 percent fitted-value band: 15 countries, compared with 19 in the World Bank regional model, 25 in the WHO regional model, and 28 in the no-region model. This supports the view that regional variables improve the fit and that UN M49 is the strongest of the current regional specifications.

The coefficient tables for the UN M49, WHO, World Bank, and no-region specifications are useful for comparing model outputs across alternative specifications. They show whether signs, magnitudes, and statistical significance change across region definitions. However, they do not test all modelling assumptions by themselves.

The coefficient stability summary in Appendix Table 21 answers whether the same country-level variables remain important across specifications. It shows two patterns. First, urban population percentage is significant in all four specifications, with negative coefficients and p-values below 0.05. Second, World Bank income-group variables are especially important when UN M49 regions are not used: low-income status and lower-middle-income status are significant in the World Bank, WHO, and no-region models. In the UN M49 model, low-income status is weaker and does not reach the 0.05 threshold ($p = 0.0714$), suggesting that some income-related structure may be absorbed by the more detailed UN M49 regional categories.

The numerical health and development indicators are less stable across specifications. Life expectancy is significant in the UN M49 model ($p = 0.0128$) but not in the World Bank, WHO, or no-region models. WHO DTP3 immunization coverage is significant only in the no-region model ($p = 0.0152$). GDP per capita, GNI per capita, physician density, under-five mortality, maternal mortality, and internet-user percentage do not remain statistically significant in the multivariable coefficient tables, even though several of them have strong bivariate correlations with approval rate. This difference is consistent with overlapping information among country-level development indicators.

This does not mean that these parameters are substantively unimportant. Their bivariate correlations show that they are part of the broader country-level pattern associated with approval rates. In the multivariable models, however, several indicators describe related dimensions of national income, health systems, infrastructure, and institutional development. Once these variables are entered together with income groups and regional classifications, their separate coefficients become harder to distinguish statistically. Their substantive importance is therefore better understood as part of a shared country-level development profile, rather than as isolated independent predictors.

The residual summaries and residuals-versus-fitted plots provide an additional model check. Across all four specifications, mean residuals are 0.0 after rounding, which is expected for these fitted linear models. Table 12 shows that residual spread follows the same ordering as the score tables: the UN M49 model has the smallest residual standard deviation and mean absolute residual, while the no-region model has the largest values. Figure 3 shows the same comparison graphically. This supports the same conclusion as the score and prediction-rate checks: the UN M49 specification leaves less unexplained error than the alternatives.

Table 12. Residual summary by regression specification

Specification	Residual SD	MAE	Max. abs. residual
UN M49 regional model	0.090	0.069	0.294
World Bank regional model	0.108	0.082	0.314
WHO regional model	0.115	0.087	0.384
No-region model	0.134	0.100	0.437

The largest positive and negative relative residuals are reported in Appendix Tables 22 and 23. These tables use `Prediction_rate`, the ratio of observed approval rate to fitted approval rate, rather than only the absolute residual. This is useful because the same absolute difference can be more consequential when the fitted approval rate is low. The positive table identifies countries where observed approval is high relative to the fitted value; the negative table identifies countries where observed approval is low relative to the fitted value.

The relative-residual tables also show that several countries recur among the largest positive or negative deviations across model specifications. This supports two interpretations. First, the model comparison is stable enough to support the conclusion that the UN M49 specification has the best fit among the current alternatives. Second, the largest country-level deviations are not explained only by the choice of regional classification. Persistent deviations across specifications may indicate omitted country-specific conditions, applicant-composition differences, documentation or verification factors, or other application-level mechanisms not represented in the current country-level dataset.

An additional weighted UN M49 experiment was estimated in Python using the base-10 logarithm of total applications as the observation weight. This gives more influence to countries with larger log-transformed application volumes while preserving the same UN M49 feature structure. Table 13 shows that the weighted model remains close to the unweighted UN M49 model. The mean signed error rounds to 0.0 in all three evaluations. These results indicate that the main UN M49 fit result is stable when countries with larger log-transformed application volumes receive more weight.

The weighted coefficient table also supports a cautious interpretation. Table 14 shows that life expectancy at birth and urban population percentage remain statistically significant in the weighted UN M49 model. Other variables that were significant in the unweighted UN M49 model, including several regional indicators and the missing World Bank lending-category term, no longer cross the 0.05 threshold. This pattern is substantively plausible. Weighting by

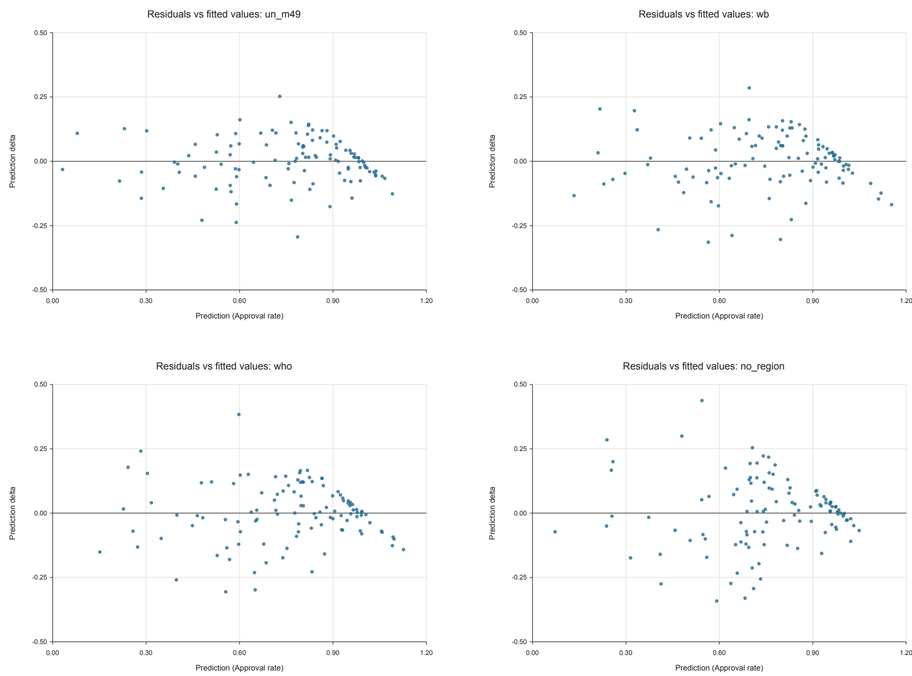

Fig. 3. Residuals against fitted approval rates in the four regression specifications. The figures are generated from `data/support/residuals_vs_fitted_un_m49.svg`, `data/support/residuals_vs_fitted_wb.svg`, `data/support/residuals_vs_fitted_who.svg`, and `data/support/residuals_vs_fitted_no_region.svg`.

Table 13. Weighted UN M49 model fit

Evaluation	R-squared	MAE	RMSE
Weighted model, weighted metrics	0.855	0.068	0.088
Weighted model, unweighted metrics	0.864	0.071	0.091
Main UN M49 model, unweighted metrics	0.867	0.069	0.090

application volume reduces the statistical prominence of several regional terms, suggesting that some region-level coefficient patterns in the unweighted model are influenced by treating each country equally regardless of application volume. When countries with more applications receive more weight, the overall UN M49 model fit remains similar, but the evidence for specific regional coefficients becomes weaker. The weighted experiment therefore strengthens the model-fit conclusion while reinforcing caution about individual coefficients.

Table 14. Terms below $p < 0.05$ in the weighted UN M49 specification

Term	Coefficient	p
Life expectancy at birth	0.543	0.0078
Urban population percentage	-0.269	0.0051

These checks do not turn the analysis into a causal design. Their role is narrower: to show whether the descriptive model fit and the comparison between regional classifications are stable under reasonable alternative specifications.

7 Discussion

The results show that country-level variation in INZ fee-paying student visa approval rates is structured rather than random. Application volume has a positive but limited association with approval rate, as shown in Figure 1. This may reflect application ecosystems that develop around high-volume countries: education agents, advisers, institutional experience, previous successful applicants, and better knowledge of documentary expectations. The present data do not measure these mechanisms directly, so this explanation should be treated as an interpretation rather than a tested cause.

The joined WHO, World Bank, and UN M49 data show that approval rates are also associated with broad country-level conditions. Countries with higher life expectancy, higher internet-use rates, higher physician density, and higher income measures tend to have higher approval rates, while countries with higher mortality indicators tend to have lower approval rates; the strongest bivariate correlations are reported in Table 3. These variables should not be read as direct determinants of individual decisions. They are better understood as indicators of the administrative, economic, health, and institutional context from which applicants prepare evidence for a formal visa process.

The regression results support the same general interpretation. The no-region model already explains a substantial share of the variation, which means that non-region country-level variables contain important information. Adding regional classification improves model fit, and the UN M49 specification performs best among the current alternatives, as shown in Table 5 and Figure 2. This suggests that regional structure matters, but it does not mean that region itself is a causal factor. Regional classifications are summary categories that may capture shared institutional, economic, mobility, documentation, and applicant-composition conditions that are not directly observed in the dataset.

The weighted UN M49 experiment adds an important qualification. When countries with larger log-transformed application volumes receive more weight, the overall model fit remains close to the unweighted UN M49 model, but several regional terms and the missing World Bank lending-category term no longer cross the $p < 0.05$ threshold, as shown in Tables 13 and 14. This supports the main fit conclusion while reducing confidence in strong interpretations of individual regional coefficients. Some regional patterns are more visible when each country is treated equally, and weaker when high-volume applicant countries receive more influence.

The residual and relative-residual checks show where the current model does not explain observed approval rates well. Positive deviations, where observed approval is higher than fitted approval, may reflect country-specific conditions not captured by the model; these cases are visible in Figure 3 and Appendix Table 22. China is a useful example because it has the largest application volume in the prepared INZ table, with 35,472 applications, but its observed approval rate is still higher than the fitted value in the UN M49 model: 0.966 observed compared with 0.822 fitted, as shown in Appendix Table 24. Possible examples include inter-governmental or inter-educational relationships, established pathways between education providers, scholarship or sponsorship channels, or unusually effective application support structures. Work on China Scholarship Council programmes illustrates how formal scholarship channels and international agreements can structure academic mobility [40]. These possibilities are not measured in the current data, but they identify plausible directions for further research.

Negative deviations require even more caution. Countries with observed approval rates well below fitted values cannot be explained by the current country-level predictors alone; the largest negative relative residuals are reported in Appendix Table 23. These cases may involve applicant-composition differences, programme choice, documentation problems, financial-evidence issues, verification difficulties, or other application-level factors that are absent from the dataset. They should therefore be treated as questions for further empirical investigation, not as evidence for a specific institutional mechanism.

Overall, the analysis supports a narrow conclusion: country-level approval-rate differences are associated with application volume, national development indicators, income categories, and regional classification systems. This conclusion is based on the descriptive application-volume pattern, the bivariate correlations, and the model comparisons reported in Figure 1, Table 3, and Table 5. Within the interpretive framework of this article, these differences are treated as possible evidence of different applicant conditions across countries and regions under the same formal decision framework. The results are useful for describing and modelling variation at the country level, but they do not identify the individual-level reasons why an application is approved or declined.

8 Limitations

The analysis is limited by aggregation at the country level. The unit of observation is country or applicant nationality, not an individual application. The models therefore cannot separate applicant-level characteristics, institutional documentation quality, programme choice, education provider characteristics, or officer-level assessment details.

The country-level predictors also overlap substantially. Table 3 shows strong pairwise correlations among health, income, internet-use, mortality, and life-expectancy indicators. This supports the substantive interpretation that these variables describe a shared country-level development profile, but it limits interpretation of individual regression coefficients. A full variance inflation factor analysis or other design-matrix diagnostic could examine this more formally, but the present article treats the existing correlation output as sufficient evidence that individual coefficients should be interpreted cautiously.

The results are also sensitive to country matching, missing numerical values, regional classification choices, and the exclusion of countries with fewer than 10 total applications across 2022–2025. Appendix Table 16 reports the excluded countries, and Table 4 reports the regional classification alternatives used in the model comparison. The comparison across UN M49, World Bank, WHO, and no-region specifications addresses part of this sensitivity, but it does not remove the broader limitations of observational, aggregated data.

9 Conclusion

This article shows that country-level variation in Immigration New Zealand fee-paying student visa approval rates during 2022–2025 is associated with application volume, country-level development indicators, income categories, and regional classification systems. The strongest current model fit is obtained with the UN M49 regional specification, as reported in Table 5, but the results also show that non-region country-level variables carry substantial explanatory information. The analysis should therefore be read as a country-level descriptive and modelling exercise, not as a causal account of individual visa decisions or as evidence about officer-level assessment behaviour.

The main empirical contribution is to connect INZ student visa decision outcomes with external country-level reference data in a reproducible workflow. This makes visible both broad structure and unexplained country-specific deviations. The broad structure is summarized in Tables 3 and 5; the deviations are summarized in Tables 10 and 11. Together, these outputs support cautious interpretation of country-level application conditions and identify where the current data are insufficient and where more detailed research is needed.

It is difficult to assume that WHO and World Bank indicators directly influence visa approval rates. A more cautious interpretation is that these indicators mark broader country-specific conditions that are not directly observed in the present dataset, consistent with the correlation and coefficient-stability results in Tables 3 and 21. Such hidden conditions may influence both the measured WHO and World Bank variables and the practical circumstances under which applicants prepare visa evidence. Identifying these country-specific mechanisms is therefore a central task for further research.

Future work should extend the comparison beyond New Zealand by examining student visa decisions in other receiving countries, especially where approval, refusal, enrolment, or policy-restrictiveness data can be linked over time. Studies of the United States and Canada show that visa restrictiveness and approval patterns can affect international student enrolment and mobility [6, 9]. A comparative design would therefore help distinguish patterns that are specific to INZ from broader features of student visa systems in destination countries.

Further models should add country-level variables suggested by the student-mobility literature, provided that their definitions and provenance are clear. Relevant candidates include passport or visa-access measures linked to the global mobility divide [31, 24, 23, 10]; language, geographic distance, existing student or migrant networks, institutional quality, political freedom, fees, and origin- or destination-country economic conditions [29, 1, 3]; and formal education or development channels such as foreign aid projects, scholarship programmes, and structured inter-country education relationships [22, 40]. These variables would test whether the countries with approval rates above or below the fitted values, reported in Tables 11, 22, and 23, are associated with documented mobility channels, application-support structures, or information conditions rather than with region or development indicators alone.

To inform our empirical analysis, we propose a basic theoretical model to understand how incomplete information affects the processing of visa applications by immigration officers.

10 Appendices or Supplementary Material

The appendix tables and figures refer to repository-relative files in the project repository [18].

10.1 Repository Structure

The repository separates source data, prepared data, model results, and support outputs so that the empirical workflow can be inspected without mixing raw inputs with derived files. All paths mentioned in this article are relative to the repository root.

`data/original/` contains source files downloaded from INZ and from reference-data providers. These files preserve the input record and are not edited manually.

`data/prepared/` contains cleaned, harmonised, aggregated, and joined datasets. These files are used as intermediate research tables and as modelling inputs after country-name harmonisation and filtering.

`data/results/` contains outputs produced by the modelling workflow, including coefficient tables, score tables, fitted-value tables, prediction-rate tables, correlation outputs, and source figure files used for the main results.

`data/support/` contains derived diagnostic summaries and auxiliary tables used to interpret or document the modelling results. These include excluded-country records, residual summaries, coefficient-stability summaries, and source figure files for robustness and residual diagnostics.

10.2 Student Visa Application Preparation

The INZ student visa decision data are useful for this analysis partly because a fee-paying student visa application is unlikely to be a casual or spontaneous action. Before an application can be assessed, the applicant normally has to assemble educational, financial, personal, health, and documentary evidence. This does not prove that every application is strong, complete, or equally supported. It does mean that the published decision records represent applications that passed through a substantial preparation process before reaching a formal decision.

Table 15. Typical preparation requirements before a fee-paying student visa decision

Preparation area	Examples of required or commonly relevant evidence
Education pathway	Identifying an education provider, submitting documents to that provider, receiving an offer of place, and documenting previous education.
Tuition and insurance	Paying tuition or documenting tuition arrangements, receiving confirmation from the education provider, and arranging medical or travel insurance where required.
Financial capacity	Providing bank statements or other evidence of funds for study and living costs, together with explanations of source of funds and relevant incoming transfers.
Personal and travel history	Providing previous education and work history, explaining gaps in education or employment, and documenting travel history where requested.
Health and character	Providing relevant health information, medical examination material where required, and character-related information required by the visa process.
Home-country ties and translations	Documenting family relationships, property or other ties where relevant, and providing certified translations for documents not originally in English.

This preparation burden matters for interpretation. The observed INZ application counts are not treated as a simple measure of abstract interest in New Zealand study. They are better understood as counts of completed applications that were submitted after applicants had already undertaken a costly and document-intensive process. For this reason, variation in approval rates may reflect differences in the practical conditions under which applicants can assemble convincing evidence, not only differences in demand for study.

10.3 Countries Excluded Before Regression

Table 16 reports countries present in the aggregated INZ decision data but not retained in the final regression table. Eight countries were not retained after the reference-data join or missing-value filter. The remaining countries were excluded because they had fewer than 10 total applications across 2022–2025.

Table 16: Countries excluded before regression modelling

Country	Applications	Approval rate	Exclusion reason
Hong Kong	1397	0.981	Not retained after reference join or missing-value filter
Taiwan	1094	0.976	Not retained after reference join or missing-value filter
Macau	69	0.928	Not retained after reference join or missing-value filter
Tuvalu	59	0.983	Not retained after reference join or missing-value filter
Palestine	15	0.533	Not retained after reference join or missing-value filter
Lithuania	10	0.900	Not retained after reference join or missing-value filter
Malawi	10	0.600	Not retained after reference join or missing-value filter
Trinidad and Tobago	10	0.800	Not retained after reference join or missing-value filter
Iceland	9	1.000	Fewer than 10 total applications
Liberia	9	0.000	Fewer than 10 total applications
Namibia	9	0.556	Fewer than 10 total applications
Guatemala	8	0.750	Fewer than 10 total applications
Nauru	8	1.000	Fewer than 10 total applications
Albania	7	0.714	Fewer than 10 total applications
Benin	7	0.286	Fewer than 10 total applications
Estonia	7	1.000	Fewer than 10 total applications
Georgia	7	0.571	Fewer than 10 total applications
Croatia	6	1.000	Fewer than 10 total applications
El Salvador	6	1.000	Fewer than 10 total applications
Guyana	6	0.833	Fewer than 10 total applications
Marshall Islands	6	1.000	Fewer than 10 total applications
South Sudan	6	0.333	Fewer than 10 total applications
St Lucia	6	1.000	Fewer than 10 total applications
Azerbaijan	5	0.800	Fewer than 10 total applications
Gambia	5	0.200	Fewer than 10 total applications
Haiti	5	0.200	Fewer than 10 total applications
Mozambique	5	0.800	Fewer than 10 total applications
Palau	5	1.000	Fewer than 10 total applications
Suriname	5	1.000	Fewer than 10 total applications
Swaziland	5	0.800	Fewer than 10 total applications
Tunisia	5	0.400	Fewer than 10 total applications
Armenia	4	0.500	Fewer than 10 total applications
Bahrain	4	1.000	Fewer than 10 total applications
Belize	4	1.000	Fewer than 10 total applications
Burundi	4	0.000	Fewer than 10 total applications
Eritrea	4	0.750	Fewer than 10 total applications

Table 16 – continued from previous page

Country	Applications	Approval rate	Exclusion reason
Grenada	4	0.750	Fewer than 10 total applications
Macedonia	4	1.000	Fewer than 10 total applications
Mali	4	0.750	Fewer than 10 total applications
Moldova	4	0.750	Fewer than 10 total applications
Serbia	4	0.750	Fewer than 10 total applications
Sierra Leone	4	0.000	Fewer than 10 total applications
Angola	3	0.000	Fewer than 10 total applications
Barbados	3	1.000	Fewer than 10 total applications
Congo	3	0.000	Fewer than 10 total applications
Dominica	3	0.667	Fewer than 10 total applications
Federated States of Micronesia	3	1.000	Fewer than 10 total applications
Liechtenstein	3	1.000	Fewer than 10 total applications
Malta	3	1.000	Fewer than 10 total applications
Slovenia	3	1.000	Fewer than 10 total applications
Somalia	3	0.000	Fewer than 10 total applications
St Kitts - Nevis	3	0.667	Fewer than 10 total applications
St Vincent and the Grenadines	3	1.000	Fewer than 10 total applications
Tajikistan	3	0.667	Fewer than 10 total applications
Australia	2	1.000	Fewer than 10 total applications
Bermuda	2	0.500	Fewer than 10 total applications
Guinea	2	0.000	Fewer than 10 total applications
Senegal	2	0.000	Fewer than 10 total applications
Seychelles	2	1.000	Fewer than 10 total applications
Bahamas	1	1.000	Fewer than 10 total applications
Bosnia & Herzegovina	1	1.000	Fewer than 10 total applications
Cape Verde	1	1.000	Fewer than 10 total applications
Cuba	1	1.000	Fewer than 10 total applications
Djibouti	1	0.000	Fewer than 10 total applications
Madagascar	1	0.000	Fewer than 10 total applications
Nicaragua	1	1.000	Fewer than 10 total applications
Niger	1	0.000	Fewer than 10 total applications
North Korea	1	0.000	Fewer than 10 total applications
Togo	1	0.000	Fewer than 10 total applications

The following tables report the full coefficient and p-value outputs for the four regression specifications discussed in Section 5.3. The source files are `data/results/Coefficients_un_m49.csv`, `data/results/Coefficients_wb.csv`, `data/results/Coefficients_who.csv`, and `data/results/Coefficients_no_region.csv`.

Table 17: Full coefficient table for the UN M49 regional specification

Term	Coefficient	p
Urban_population_pct	-0.2795	0.0053
UN M49=Northern Africa	-0.2537	0.0109
Life_expectancy_at_birth_years	0.4992	0.0128
WB_FY2026_Lending_Category=Missing	0.2425	0.0156
UN M49=Middle Africa	-0.2749	0.0219
UN M49=Southern Asia	-0.2100	0.0336
Intercept	0.4674	0.0492
WB_FY2026_Income_Group=Low income	-0.1980	0.0714
UN M49=Melanesia	0.2055	0.0861
UN M49=Central Asia	-0.1989	0.0902
UN M49=Western Asia	-0.1674	0.0904
WB_FY2026_Lending_Category=IBRD	0.1036	0.1439
UN M49=South-eastern Asia	0.1074	0.2368

Table 17 – continued from previous page

Term	Coefficient	p
WB_FY2026_Lending_Category=IDA	0.0600	0.3012
UN M49=Micronesia	0.1634	0.3124
Internet_users_pct_population	0.1136	0.3553
UN M49=Southern Europe	-0.0925	0.4785
UN M49=Polynesia	0.0970	0.4848
GDP_per_capita_current_USD	-0.1922	0.4888
UN M49=Northern Europe	-0.0755	0.5001
GNI_per_capita_Atlas_current_USD	0.1768	0.5223
UN M49=Central America	0.0613	0.5712
WB_FY2026_Income_Group=Lower middle income	-0.0389	0.5992
UN M49=Eastern Africa	-0.0478	0.6506
UN M49=South America	0.0438	0.6554
WHO_DTP3_Immunization_Coverage_pct	-0.0355	0.6678
UN M49=Western Europe	-0.0315	0.7865
WB_FY2026_Income_Group=Missing	-0.0290	0.7943
UN M49=Northern America	-0.0335	0.7951
WB_FY2026_Income_Group=Upper middle income	-0.0148	0.7967
UN M49=Eastern Asia	-0.0261	0.8053
UN M49=Eastern Europe	0.0192	0.8582
UN M49=Southern Africa	0.0204	0.8628
WHO_Physician_Density_per_10000_population	-0.0140	0.9018
WHO_Under5_Mortality_per_1000_live_births	0.0245	0.9164
UN M49=Western Africa	-0.0105	0.9384
WHO_Maternal_Mortality_per_100000_live_births	-0.0081	0.9689

Table 18: Full coefficient table for the World Bank regional specification

Term	Coefficient	p
WB_FY2026_Region=Middle East, North Africa, Afghanistan & Pakistan	-0.2695	$8.194e{-8}$
Intercept	0.9166	$5.844e{-7}$
WB_FY2026_Income_Group=Low income	-0.4067	$5.477e{-5}$
WB_FY2026_Region=Europe & Central Asia	-0.1846	$3.689e{-4}$
WB_FY2026_Region=South Asia	-0.2322	$5.548e{-4}$
WB_FY2026_Region=Sub-Saharan Africa	-0.1561	0.0042
Urban_population_pct	-0.2545	0.0061
WB_FY2026_Lending_Category=Missing	0.2244	0.0171
WB_FY2026_Income_Group=Lower middle income	-0.1355	0.0461
WB_FY2026_Income_Group=Upper middle income	-0.0914	0.0889
WB_FY2026_Lending_Category=IBRD	0.1114	0.0918
WB_FY2026_Region=North America	-0.1633	0.1001
Life_expectancy_at_birth_years	0.2595	0.1672
WB_FY2026_Region=Latin America & Caribbean	-0.0644	0.2041
WHO_DTP3_Immunization_Coverage_pct	-0.0970	0.2097
WB_FY2026_Lending_Category=IDA	0.0692	0.2267
WB_FY2026_Income_Group=Missing	-0.1243	0.2511
WHO_Under5_Mortality_per_1000_live_births	-0.1944	0.3301
GNI_per_capita_Atlas_current_USD	0.1541	0.5663
GDP_per_capita_current_USD	-0.1087	0.7076
Internet_users_pct_population	0.0271	0.7942
WHO_Maternal_Mortality_per_100000_live_births	-0.0367	0.8419
WHO_Physician_Density_per_10000_population	-0.0041	0.9680

Table 19: Full coefficient table for the WHO regional specification

Term	Coefficient	p
Intercept	0.8794	$3.670e{-6}$
WB_FY2026_Income_Group=Low income	-0.4423	$4.232e{-5}$
WHO_Region=Western Pacific Region	0.1980	$9.745e{-4}$
WB_FY2026_Income_Group=Lower middle income	-0.1780	0.0127
Urban_population_pct	-0.2370	0.0132
WB_FY2026_Lending_Category=Missing	0.2311	0.0198
WB_FY2026_Income_Group=Upper middle income	-0.1039	0.0680
WHO_Region=Region of the Americas	0.1219	0.0691
WB_FY2026_Lending_Category=IBRD	0.1173	0.0952
WHO_DTP3_Immunization_Coverage_pct	-0.1333	0.1146
WHO_Region=South-East Asia Region	0.0849	0.1933
WB_FY2026_Income_Group=Missing	-0.1484	0.1958
WHO_Under5_Mortality_per_1000_live_births	-0.2102	0.3220
WHO_Region=Eastern Mediterranean Region	-0.0545	0.3817
WB_FY2026_Lending_Category=IDA	0.0481	0.4213
Life_expectancy_at_birth_years	0.1354	0.4701
GNI_per_capita_Atlas_current_USD	0.1327	0.6318
WHO_Region=European Region	0.0319	0.6626
WHO_Physician_Density_per_10000_population	-0.0393	0.7146
GDP_per_capita_current_USD	-0.0843	0.7816
WHO_Maternal_Mortality_per_100000_live_births	-0.0494	0.7995
Internet_users_pct_population	-0.0107	0.9230

Table 20: Full coefficient table for the no-region specification

Term	Coefficient	p
Intercept	1.1160	$3.208e{-8}$
WB_FY2026_Income_Group=Low income	-0.6087	$2.500e{-7}$
WB_FY2026_Income_Group=Lower middle income	-0.2242	0.0047
Urban_population_pct	-0.2635	0.0081
WHO_DTP3_Immunization_Coverage_pct	-0.2036	0.0152
WHO_Under5_Mortality_per_1000_live_births	-0.4155	0.0758
WB_FY2026_Income_Group=Upper middle income	-0.1059	0.0958
WB_FY2026_Income_Group=Missing	-0.2087	0.1085
WB_FY2026_Lending_Category=Missing	0.1644	0.1301
WB_FY2026_Lending_Category=IDA	0.0833	0.2166
WB_FY2026_Lending_Category=IBRD	0.0930	0.2363
WHO_Physician_Density_per_10000_population	-0.1130	0.2560
GNI_per_capita_Atlas_current_USD	0.2416	0.4372
Life_expectancy_at_birth_years	0.0871	0.6543
GDP_per_capita_current_USD	-0.1465	0.6644
Internet_users_pct_population	0.0134	0.9087
WHO_Maternal_Mortality_per_100000_live_births	0.0182	0.9329

10.4 Coefficient Stability Summary

Table 21 reports the cross-specification coefficient stability summary from `data/support/Coefficients_stabi` The table gives the number of model specifications in which each term appears, the number of those specifications where $p < 0.05$, and the minimum and maximum p-values observed across the specifications where the term is present.

Table 21: Coefficient stability summary across regression specifications

Term	Type	Models	Sig.	Min. p	Max. p	Stability
Urban_population_pct	Variable	4	4	0.0053	0.0132	stable significant
WB_FY2026_Income_Group=Low income	Variable	4	3	$2.500e{-7}$	0.0714	partly stable
WB_FY2026_Income_Group=Lower middle income	Variable	4	3	0.0047	0.5992	partly stable
WB_FY2026_Lending_Category=Missing	Variable	4	3	0.0156	0.1301	partly stable
Life_expectancy_at_birth_years	Variable	4	1	0.0128	0.6543	specification dependent
WHO_DTP3_Immunization_Coverage_pct	Variable	4	1	0.0152	0.6678	specification dependent
WB_FY2026_Income_Group=Upper middle income	Variable	4	0	0.0680	0.7967	not significant
WHO_Under5_Mortality_per_1000_live_births	Variable	4	0	0.0758	0.9164	not significant
WB_FY2026_Lending_Category=IBRD	Variable	4	0	0.0918	0.2363	not significant
WB_FY2026_Income_Group=Missing	Variable	4	0	0.1085	0.7943	not significant
WB_FY2026_Lending_Category=IDA	Variable	4	0	0.2166	0.4213	not significant
WHO_Physician_Density_per_10000_population	Variable	4	0	0.2560	0.9680	not significant
Internet_users_pct_population	Variable	4	0	0.3553	0.9230	not significant
GNI_per_capita_Atlas_current_USD	Variable	4	0	0.4372	0.6318	not significant
GDP_per_capita_current_USD	Variable	4	0	0.4888	0.7816	not significant
WHO_Maternal_Mortality_per_100000_live_births	Variable	4	0	0.7995	0.9689	not significant
WB_FY2026_Region=Middle East, North Africa, Afghanistan & Pakistan	Region	1	1	$8.194e{-8}$	$8.194e{-8}$	stable significant
WB_FY2026_Region=Europe & Central Asia	Region	1	1	$3.689e{-4}$	$3.689e{-4}$	stable significant
WB_FY2026_Region=South Asia	Region	1	1	$5.548e{-4}$	$5.548e{-4}$	stable significant
WHO_Region=Western Pacific Region	Region	1	1	$9.745e{-4}$	$9.745e{-4}$	stable significant
WB_FY2026_Region=Sub-Saharan Africa	Region	1	1	0.0042	0.0042	stable significant
UN M49=Northern Africa	Region	1	1	0.0109	0.0109	stable significant
UN M49=Middle Africa	Region	1	1	0.0219	0.0219	stable significant
UN M49=Southern Asia	Region	1	1	0.0336	0.0336	stable significant
WHO_Region=Region of the Americas	Region	1	0	0.0691	0.0691	not significant
UN M49=Melanesia	Region	1	0	0.0861	0.0861	not significant
UN M49=Central Asia	Region	1	0	0.0902	0.0902	not significant
UN M49=Western Asia	Region	1	0	0.0904	0.0904	not significant
WB_FY2026_Region=North America	Region	1	0	0.1001	0.1001	not significant
WHO_Region=South-East Asia Region	Region	1	0	0.1933	0.1933	not significant
WB_FY2026_Region=Latin America & Caribbean	Region	1	0	0.2041	0.2041	not significant
UN M49=South-eastern Asia	Region	1	0	0.2368	0.2368	not significant
UN M49=Micronesia	Region	1	0	0.3124	0.3124	not significant
WHO_Region=Eastern Mediterranean Region	Region	1	0	0.3817	0.3817	not significant
UN M49=Southern Europe	Region	1	0	0.4785	0.4785	not significant
UN M49=Polynesia	Region	1	0	0.4848	0.4848	not significant
UN M49=Northern Europe	Region	1	0	0.5001	0.5001	not significant
UN M49=Central America	Region	1	0	0.5712	0.5712	not significant
UN M49=Eastern Africa	Region	1	0	0.6506	0.6506	not significant
UN M49=South America	Region	1	0	0.6554	0.6554	not significant
WHO_Region=European Region	Region	1	0	0.6626	0.6626	not significant
UN M49=Western Europe	Region	1	0	0.7865	0.7865	not significant
UN M49=Northern America	Region	1	0	0.7951	0.7951	not significant
UN M49=Eastern Asia	Region	1	0	0.8053	0.8053	not significant
UN M49=Eastern Europe	Region	1	0	0.8582	0.8582	not significant
UN M49=Southern Africa	Region	1	0	0.8628	0.8628	not significant
UN M49=Western Africa	Region	1	0	0.9384	0.9384	not significant

10.5 Relative Residual Tables

Tables 22 and 23 report the largest positive and negative relative residuals by model specification. Each entry gives the country, `Prediction_rate`, and residual delta. The source files are `data/support/Relative_residuals_positive_by_model.csv` and `data/support/Relative_residuals_negative_by_model.csv`.

Table 22: Largest positive relative residuals by model specification

UN M49	World Bank	WHO	No region
Democratic Republic of Congo (rate=2.375, delta=0.109)	Nigeria (rate=1.940, delta=0.204)	Rwanda (rate=1.852, delta=0.241)	Rwanda (rate=2.191, delta=0.285)

Table 22 – continued from previous page

UN M49	World Bank	WHO	No region
Sudan (rate=1.551, delta=0.127)	Rwanda (rate=1.601, delta=0.197)	Nigeria (rate=1.736, delta=0.178)	Timor Leste (rate=1.804, delta=0.437)
Nigeria (rate=1.392, delta=0.118)	Timor Leste (rate=1.410, delta=0.285)	Timor Leste (rate=1.642, delta=0.384)	Uganda (rate=1.774, delta=0.200)
Timor Leste (rate=1.346, delta=0.252)	Uganda (rate=1.364, delta=0.122)	Uganda (rate=1.508, delta=0.154)	Nigeria (rate=1.659, delta=0.167)
Kazakhstan (rate=1.269, delta=0.161)	Botswana (rate=1.242, delta=0.146)	Jordan (rate=1.247, delta=0.118)	Kiribati (rate=1.624, delta=0.299)
Jamaica (rate=1.198, delta=0.151)	Ukraine (rate=1.233, delta=0.162)	Botswana (rate=1.245, delta=0.148)	Laos (rate=1.360, delta=0.254)
Zambia (rate=1.195, delta=0.103)	Lebanon (rate=1.213, delta=0.122)	Kiribati (rate=1.240, delta=0.151)	Solomon Islands (rate=1.300, delta=0.222)
Lebanon (rate=1.184, delta=0.108)	Kiribati (rate=1.202, delta=0.131)	Zambia (rate=1.238, delta=0.121)	Malaysia (rate=1.287, delta=0.218)
China (rate=1.175, delta=0.144)	Laos (rate=1.196, delta=0.157)	Laos (rate=1.207, delta=0.165)	Fiji (rate=1.282, delta=0.175)
Bolivia (rate=1.171, delta=0.121)	Kuwait (rate=1.185, delta=0.154)	Kuwait (rate=1.203, delta=0.166)	Brazil (rate=1.276, delta=0.193)
Laos (rate=1.169, delta=0.139)	Jordan (rate=1.179, delta=0.090)	Indonesia (rate=1.198, delta=0.157)	Cambodia (rate=1.270, delta=0.195)
Sri Lanka (rate=1.164, delta=0.109)	Brazil (rate=1.176, delta=0.134)	Lebanon (rate=1.198, delta=0.115)	China (rate=1.240, delta=0.187)
Maldives (rate=1.154, delta=0.110)	Jamaica (rate=1.171, delta=0.134)	Ukraine (rate=1.198, delta=0.141)	Jamaica (rate=1.206, delta=0.157)
Oman (rate=1.146, delta=0.122)	Poland (rate=1.166, delta=0.143)	Brazil (rate=1.192, delta=0.143)	Philippines (rate=1.198, delta=0.138)
Rwanda (rate=1.144, delta=0.066)	Zambia (rate=1.165, delta=0.090)	Solomon Islands (rate=1.168, delta=0.139)	Mexico (rate=1.196, delta=0.151)

Table 23: Largest negative relative residuals by model specification

UN M49	World Bank	WHO	No region
Chad (rate=0.000, delta=-0.032)	Chad (rate=0.000, delta=-0.133)	Chad (rate=0.000, delta=-0.151)	Chad (rate=0.000, delta=-0.073)
Afghanistan (rate=0.495, delta=-0.144)	Cameroon (rate=0.342, delta=-0.266)	Cameroon (rate=0.347, delta=-0.259)	Cameroon (rate=0.334, delta=-0.275)
Uzbekistan (rate=0.522, delta=-0.229)	Uzbekistan (rate=0.443, delta=-0.314)	Uzbekistan (rate=0.450, delta=-0.306)	Uzbekistan (rate=0.423, delta=-0.342)
Turkey (rate=0.597, delta=-0.237)	Turkey (rate=0.550, delta=-0.288)	Afghanistan (rate=0.517, delta=-0.132)	Afghanistan (rate=0.447, delta=-0.174)
Myanmar (rate=0.626, delta=-0.294)	Afghanistan (rate=0.615, delta=-0.088)	Turkey (rate=0.541, delta=-0.298)	Turkey (rate=0.516, delta=-0.330)
Cameroon (rate=0.642, delta=-0.077)	Myanmar (rate=0.619, delta=-0.304)	Algeria (rate=0.643, delta=-0.231)	Libya (rate=0.571, delta=-0.273)
Yemen (rate=0.704, delta=-0.105)	Tanzania (rate=0.710, delta=-0.173)	Nepal (rate=0.683, delta=-0.180)	Algeria (rate=0.587, delta=-0.293)
Tanzania (rate=0.719, delta=-0.166)	Algeria (rate=0.726, delta=-0.157)	Libya (rate=0.688, delta=-0.165)	Yemen (rate=0.610, delta=-0.160)

Table 23 – continued from previous page

UN M49	World Bank	WHO	No region
Algeria (rate=0.794, delta=-0.108)	Democratic Republic of Congo (rate=0.727, delta=-0.070)	Yemen (rate=0.718, delta=-0.098)	Tanzania (rate=0.645, delta=-0.233)
Ghana (rate=0.794, delta=-0.118)	Mongolia (rate=0.727, delta=-0.227)	Myanmar (rate=0.719, delta=-0.193)	Iraq (rate=0.650, delta=-0.256)
Ecuador (rate=0.802, delta=-0.176)	Libya (rate=0.749, delta=-0.122)	Mongolia (rate=0.726, delta=-0.228)	Nepal (rate=0.693, delta=-0.172)
Dominican Republic (rate=0.803, delta=-0.151)	Dominican Republic (rate=0.810, delta=-0.145)	Democratic Republic of Congo (rate=0.727, delta=-0.070)	Myanmar (rate=0.698, delta=-0.213)
Iraq (rate=0.835, delta=-0.094)	Ecuador (rate=0.814, delta=-0.163)	Tanzania (rate=0.759, delta=-0.135)	Iran (rate=0.730, delta=-0.197)
Vietnam (rate=0.851, delta=-0.143)	Nepal (rate=0.827, delta=-0.081)	India (rate=0.765, delta=-0.173)	Democratic Republic of Congo (rate=0.789, delta=-0.050)
Syria (rate=0.853, delta=-0.042)	Yemen (rate=0.842, delta=-0.047)	Iraq (rate=0.797, delta=-0.121)	Zimbabwe (rate=0.790, delta=-0.106)

10.6 Prediction Table

Table 24 reports the observed approval rate and the fitted approval rates from the four regression specifications for each country in the modelling table. The source files are `data/results/Prediction_un_m49.csv`, `data/results/Prediction_wb.csv`, `data/results/Prediction_who.csv`, and `data/results/Prediction_no_region.csv`.

Table 24: Observed and fitted approval rates by regression specification

Country	Observed	UN M49	WB	WHO	No region
Afghanistan	0.141	0.285	0.229	0.273	0.315
Algeria	0.417	0.525	0.574	0.648	0.710
Argentina	0.858	0.803	0.797	0.775	0.721
Austria	1.000	1.038	1.016	1.006	1.022
Bangladesh	0.558	0.587	0.605	0.678	0.669
Belarus	0.750	0.759	0.664	0.671	0.703
Belgium	0.995	0.982	0.946	0.924	0.909
Bhutan	0.632	0.572	0.588	0.656	0.669
Bolivia	0.826	0.705	0.728	0.740	0.696
Botswana	0.750	0.686	0.604	0.602	0.657
Brazil	0.892	0.781	0.758	0.748	0.699
Brunei Darussalam	1.000	1.058	1.086	1.093	0.958
Bulgaria	0.833	0.803	0.783	0.805	0.841
Cambodia	0.916	0.834	0.818	0.796	0.721
Cameroon	0.138	0.215	0.403	0.397	0.413
Canada	0.986	1.002	0.986	1.056	0.997
Chad	0.000	0.032	0.133	0.151	0.073
Chile	0.901	0.891	0.908	0.879	0.824
China	0.966	0.822	0.919	0.899	0.779
Colombia	0.748	0.836	0.806	0.790	0.741
Costa Rica	0.875	0.921	0.918	0.892	0.834
Czech Republic	0.990	1.032	0.968	0.976	0.990
Democratic Republic of Congo	0.188	0.079	0.258	0.258	0.238
Denmark	0.997	0.955	0.963	0.949	0.973
Dominican Republic	0.615	0.767	0.760	0.752	0.687
Ecuador	0.714	0.890	0.878	0.873	0.851
Egypt	0.561	0.526	0.588	0.595	0.694
Ethiopia	0.641	0.645	0.651	0.637	0.713

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Country	Observed	UN M49	WB	WHO	No region
Fiji	0.795	0.783	0.687	0.721	0.620
Finland	0.912	0.988	0.997	0.992	1.022
France	0.978	1.018	0.971	0.965	0.981
Germany	0.996	0.982	0.963	0.963	0.984
Ghana	0.455	0.573	0.516	0.464	0.555
Great Britain	0.984	0.941	0.953	0.955	0.979
Greece	0.917	0.917	0.942	0.926	0.893
Honduras	0.727	0.756	0.746	0.717	0.721
Hungary	0.982	0.983	0.919	0.934	0.942
India	0.566	0.598	0.632	0.739	0.686
Indonesia	0.950	0.859	0.870	0.793	0.820
Iran	0.531	0.590	0.594	0.603	0.727
Iraq	0.476	0.570	0.559	0.597	0.732
Ireland	0.981	1.004	1.027	1.019	1.049
Israel	0.865	0.843	0.854	0.931	0.940
Italy	0.995	0.999	1.007	0.992	1.000
Jamaica	0.917	0.765	0.783	0.787	0.760
Japan	0.996	0.994	1.119	1.096	0.942
Jordan	0.595	0.570	0.505	0.477	0.543
Kazakhstan	0.762	0.601	0.705	0.712	0.748
Kenya	0.464	0.487	0.494	0.482	0.547
Kiribati	0.779	0.779	0.648	0.628	0.479
Kuwait	0.984	0.865	0.830	0.818	0.914
Kyrgyzstan	0.667	0.599	0.683	0.655	0.740
Laos	0.960	0.821	0.803	0.795	0.706
Latvia	0.917	0.892	0.928	0.945	0.961
Lebanon	0.696	0.588	0.574	0.581	0.689
Libya	0.364	0.406	0.485	0.528	0.637
Luxembourg	1.000	0.923	0.916	0.916	0.936
Malaysia	0.976	0.910	0.878	0.869	0.758
Maldives	0.827	0.717	0.737	0.797	0.858
Mauritius	0.865	0.804	0.790	0.757	0.855
Mexico	0.923	0.818	0.803	0.802	0.772
Mongolia	0.605	0.698	0.831	0.832	0.689
Morocco	0.529	0.541	0.566	0.555	0.652
Myanmar	0.493	0.786	0.796	0.685	0.706
Nepal	0.388	0.391	0.469	0.568	0.560
Netherlands	0.990	0.959	0.933	0.931	0.955
Nigeria	0.420	0.302	0.216	0.242	0.253
Norway	0.983	1.009	1.014	0.991	1.031
Oman	0.956	0.835	0.827	0.833	0.924
Pakistan	0.391	0.402	0.379	0.399	0.458
Panama	0.919	0.949	0.984	0.988	0.974
Papua New Guinea	0.863	0.847	0.803	0.858	0.769
Paraguay	0.864	0.806	0.803	0.798	0.744
Peru	0.716	0.826	0.796	0.789	0.751
Philippines	0.838	0.822	0.824	0.840	0.700
Poland	1.000	0.902	0.857	0.865	0.913
Portugal	0.963	0.987	0.998	0.977	0.956
Romania	0.913	0.909	0.882	0.906	0.976
Russia	0.693	0.775	0.763	0.783	0.818
Rwanda	0.524	0.458	0.327	0.283	0.239
Samoa	0.955	0.880	0.940	0.957	0.927
Saudi Arabia	0.772	0.808	0.826	0.831	0.928
Singapore	0.964	1.091	1.111	1.091	0.956
Slovakia	1.000	1.066	0.987	0.993	1.001
Solomon Islands	0.963	0.912	0.834	0.825	0.741
South Africa	0.621	0.685	0.639	0.652	0.744
South Korea	0.985	1.037	1.153	1.126	0.981

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Country	Observed	UN M49	WB	WHO	No region
Spain	0.997	0.969	0.972	0.956	0.957
Sri Lanka	0.778	0.668	0.716	0.777	0.806
Sudan	0.357	0.230	0.370	0.317	0.373
Sweden	0.988	0.973	0.967	0.949	0.962
Switzerland	0.980	1.037	0.999	0.978	1.008
Syria	0.243	0.285	0.210	0.227	0.255
Tanzania	0.424	0.590	0.597	0.559	0.657
Thailand	0.925	0.949	0.916	0.805	0.827
Timor Leste	0.981	0.729	0.696	0.598	0.544
Tonga	0.863	0.937	0.944	0.928	0.895
Turkey	0.352	0.589	0.640	0.650	0.682
Uganda	0.458	0.437	0.336	0.304	0.258
Ukraine	0.857	0.789	0.695	0.716	0.760
United Arab Emirates	1.000	0.881	0.875	0.865	0.974
United States of America	0.984	0.967	0.983	1.058	1.011
Uruguay	0.828	0.812	0.866	0.845	0.783
Uzbekistan	0.250	0.479	0.564	0.556	0.592
Vanuatu	0.878	0.957	0.901	0.900	0.843
Venezuela	0.718	0.714	0.708	0.722	0.646
Vietnam	0.818	0.961	0.893	0.863	0.701
Yemen	0.250	0.355	0.297	0.348	0.410
Zambia	0.632	0.528	0.542	0.510	0.567
Zimbabwe	0.400	0.458	0.459	0.449	0.506

References

1. Abbott, A., Silles, M.: Determinants of international student migration. The World Economy **39**(5), 621–635 (2016). https://doi.org/10.1111/twec.12319, https://doi.org/10.1111/twec.12319
2. Bertoli, S., Fernández-Huertas Moraga, J.: Multilateral resistance to migration. Journal of Development Economics **102**, 79–100 (2013). https://doi.org/10.1016/j.jdeveco.2012.12.001, https://www.sciencedirect.com/science/article/pii/S0304387812001034
3. Bessey, D.: International student migration to germany. Empirical Economics **42**(1), 345–361 (2012). https://doi.org/10.1007/s00181-010-0417-0, https://doi.org/10.1007/s00181-010-0417-0
4. Blake, W.: Auguries of innocence (1863), https://www.poetryfoundation.org/poems/43650/auguries-of-innocence, in D. G. Rossetti (ed.), Poems. Retrieved from Poetry Foundation
5. Bonnett, G.: Inz's visa processing systems are 'creaking' – briefing. RNZ (Aug 2024), https://www.rnz.co.nz/news/national/523805/inz-s-visa-processing-systems-are-creaking-briefing
6. Chen, M., Howell, J., Smith, J.: Best and brightest? the impact of student visa restrictiveness on who attends college in the us. Labour Economics **84**, 102385 (2023). https://doi.org/10.1016/j.labeco.2023.102385, https://www.sciencedirect.com/science/article/pii/S092753712300060X
7. Craymer, L.: New zealand aims to double foreign international education market by 2034. Reuters (2025), https://www.reuters.com/world/asia-pacific/new-zealand-aims-double-foreign-international-education-market-by-2034-2025-07-13/
8. Education New Zealand: International student enrolments continue upward surge. Education New Zealand (2025), https://www.enz.govt.nz/news-and-research/

ed-news/international-student-enrolments-continue-upward-surge, retrieved May 12, 2026
9. Gonnot, J., Lanati, M.: Visa policy and international student migration: evidence from canada. Journal of Economic Geography **25**(2), 255–291 (2025). https://doi.org/10.1093/jeg/lbae034, https://doi.org/10.1093/jeg/lbae034
10. Gülel, D.: Navigating visa inequities: mobility as privilege in academia – 'you are not supposed to be here'. Global Social Challenges Journal **4**(2), 429–438 (2025). https://doi.org/10.1332/27523349Y2025D000000050, https://bristoluniversitypressdigital.com/gsc/view/journals/gscj/4/2/article-p429.xml
11. Hersch, J.: Profiling the new immigrant worker: The effects of skin color and height. Journal of Labor Economics **26**(2), 345–386 (2008), https://ideas.repec.org/a/ucp/jlabec/v26y2008i2p345-386.html
12. Immigration New Zealand: Offshore student visa application decisions for 2022. New Zealand Government (nd), https://www.immigration.govt.nz/study/for-education-providers/data-and-processing-times-for-international-student-visas/offshore-student-visa-application-decision-data/offshore-student-visa-application-decisions-for-2022/, retrieved May 11, 2026
13. Immigration New Zealand: Offshore student visa application decisions for 2023. New Zealand Government (nd), https://www.immigration.govt.nz/study/for-education-providers/data-and-processing-times-for-international-student-visas/offshore-student-visa-application-decision-data/offshore-student-visa-application-decisions-for-2023/, retrieved May 11, 2026
14. Immigration New Zealand: Overseas student visa application decisions for 2024. New Zealand Government (nd), https://www.immigration.govt.nz/study/for-education-providers/data-and-processing-times-for-international-student-visas/offshore-student-visa-application-decision-data/overseas-student-visa-application-decisions-for-2024/, retrieved May 11, 2026
15. Immigration New Zealand: Overseas student visa application decisions for 2025. New Zealand Government (nd), https://www.immigration.govt.nz/study/for-education-providers/data-and-processing-times-for-international-student-visas/offshore-student-visa-application-decision-data/overseas-student-visa-application-decisions-for-2025/, retrieved May 11, 2026
16. Immigration New Zealand: Providing evidence and documents to support your visa application. New Zealand Government (nd), https://www.immigration.govt.nz/process-to-apply/applying-for-a-visa/providing-evidence-and-documents-to-support-your-visa-application/, retrieved May 11, 2026
17. James, G., Witten, D., Hastie, T., Tibshirani, R.: An Introduction to Statistical Learning: With Applications in R. Springer, 2 edn. (2021). https://doi.org/10.1007/978-1-0716-1418-1, https://doi.org/10.1007/978-1-0716-1418-1

18. Kaplan, V.: Inz visa decisions research repository. GitHub repository (2026), https://github.com/vitalii-kaplan/inz-visa-decisions, retrieved May 13, 2026
19. Kaplan, V.: k2pweb.org (2026), https://k2pweb.org/, retrieved May 13, 2026
20. Kaplan, V.: knime2py. GitHub repository (2026), https://github.com/vitalii-kaplan/knime2py, retrieved May 13, 2026
21. KNIME AG: Knime analytics platform (nd), https://www.knime.com/knime-analytics-platform, retrieved May 13, 2026
22. Lanati, M., Thiele, R.: International student flows from developing countries: Do donors have an impact? Economics of Education Review **77**, 101997 (2020). https://doi.org/10.1016/j.econedurev.2020.101997, https://doi.org/10.1016/j.econedurev.2020.101997
23. Mau, S., Gülzau, F., Laube, L., Zaun, N.: The global mobility divide: How visa policies have evolved over time. Journal of Ethnic and Migration Studies **41**(8), 1192–1213 (2015). https://doi.org/10.1080/1369183X.2015.1005007, https://doi.org/10.1080/1369183X.2015.1005007
24. Neumayer, E.: Unequal access to foreign spaces: how states use visa restrictions to regulate mobility in a globalized world. Transactions of the Institute of British Geographers **31**(1), 72–84 (2006). https://doi.org/10.1111/j.1475-5661.2006.00194.x, https://doi.org/10.1111/j.1475-5661.2006.00194.x
25. New Zealand Ministry of Education: International education. New Zealand Government (2025), https://www.education.govt.nz/our-work/strategies-policies-and-programmes/tertiary-and-further-education/international-education, retrieved May 12, 2026
26. New Zealand Tertiary Education Commission: Tertiary learner snapshot 2024. New Zealand Government (2026), https://www.tec.govt.nz/sector-governance-and-performance/monitoring-and-auditing-performance/sector-performance-data/tertiary-learner-snapshot-2024, retrieved May 12, 2026
27. Oduwaye, O., Kiraz, A., Sorakin, Y.: A trend analysis of the challenges of international students over 21 years. Sage Open **13**(4) (2023). https://doi.org/10.1177/21582440231210387, https://doi.org/10.1177/21582440231210387
28. Oreopoulos, P.: Why do skilled immigrants struggle in the labor market? a field experiment with thirteen thousand resumes. American Economic Journal: Economic Policy **3**(4), 148–171 (2011). https://doi.org/10.1257/pol.3.4.148, https://doi.org/10.1257/pol.3.4.148
29. Perkins, R., Neumayer, E.: Geographies of educational mobilities: Exploring the uneven flows of international students. The Geographical Journal **180**(3), 246–259 (2014). https://doi.org/10.1111/geoj.12045, https://doi.org/10.1111/geoj.12045
30. Sassen, S.: Losing Control? Sovereignty in an Age of Globalization. Columbia University Press, New York (1996)
31. Shachar, A., Bauböck, R.: Should citizenship be for sale? EUI Working Papers RSCAS 1 (2014)
32. Spoonley, P.: Immigration challenges and changes in new zealand: Contemporary policy issues. New Zealand Journal of Employment Relations **49**(1) (2025). https://doi.org/10.24135/nzjer.v49i1.187, https://ojs.aut.ac.nz/nzjer/article/view/187

33. UNESCO: Record number of higher education students highlights global need for recognition of qualifications (Jun 2025), https://www.unesco.org/en/articles/record-number-higher-education-students-highlights-global-need-recognition-qualifications
34. United Nations Statistics Division: Standard country or area codes for statistical use (m49). United Nations (nd), https://unstats.un.org/unsd/methodology/m49/, retrieved May 11, 2026
35. Weber, T., Van Mol, C.: The student migration transition: an empirical investigation into the nexus between development and international student migration. Comparative Migration Studies **11** (2023). https://doi.org/10.1186/s40878-023-00329-0, https://doi.org/10.1186/s40878-023-00329-0
36. World Bank: World bank country and lending groups. World Bank Data Help Desk (2026), https://datahelpdesk.worldbank.org/knowledgebase/articles/906519-world-bank-country-and-lending-groups, retrieved May 11, 2026
37. World Bank: World development indicators. World Bank DataBank (2026), https://databank.worldbank.org/source/world-development-indicators, retrieved May 11, 2026
38. World Health Organization: Who data (2026), https://data.who.int/, retrieved May 11, 2026
39. World Health Organization: Global health observatory (nd), https://www.who.int/data/gho, retrieved May 11, 2026
40. Xie, Q., Yegros-Yegros, A.: A quantitative assessment of potential benefits and challenges of international researcher mobility for home and host countries: evidence from the chinese scholarship council programmes. Scientometrics **130**(8), 4545–4572 (2025). https://doi.org/10.1007/s11192-025-05391-2, https://doi.org/10.1007/s11192-025-05391-2